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Introduction

Innovation, experiential learning, multidisciplinary thinking, and technology integration into education are key features of NEP 2020. The NEP Hackathon, accordingly, offers a vibrant platform for identification of real-world challenges by students themselves and coming up with technology-driven and impactful solutions through collaboration.

This hackathon encourages participants to think critically, design creatively, and apply problem-solving skills to issues faced by educational institutions, communities, and public systems. It assembles learners from very different walks of life to ideate, prototype, and present solutions that reflect the NEP's vision for holistic, skill-oriented, innovation-driven education.

This report summarizes the entire journey of our participation in the NEP Hackathon: from problem identification, research, and solution design to prototyping, implementation, and evaluation. The report covers the different kinds of tools, methods, teamwork, and technical approaches that were followed during the hackathon and the key learnings from the process. This report will give a comprehensive overview of our project and will explain how our solution meaningfully contributes toward the mission of the NEP in transforming India's educational and societal landscape.

Purpose of the Hackathon

It additionally seeks to expose students to industry expectations, introduce them to documented novel tech mentors, and assist them in developing technical and presentation skills. Through this and subsequent events, the institution has a vision of building an innovation-centric culture that supports, as the institution grows a larger community of learners, student-driven projects that might have potential value for them as they pursue their own academic or entrepreneurial dreams.

Beyond the competition of the event itself, the hackathon also aims to build confidence, foster collaboration between students from different departments, and solidify the technical identity of ICON Fest, which recognizes an intersection of creativity in design and problem-solving.

Event Details

The CodeIcon Hackathon is a 24-hour national-level technical event organised as a part of ICON Fest by the MCA 2024-26, KJSIM, Mumbai. The hackathon is set to be held from 18-19 December 2023 on the campus to encourage innovation, collaborative problem-solving, and interdisciplinary technology development for students.

Around 50 teams, comprising about 200 participants from various academic backgrounds are taking part in the event. CodeIcon has an objective of providing an opportunity for participants to work seamlessly on real world problem statements and give working prototypes within the time allocated of the event.

The Institute will provide full logistic support including technical infrastructure, food and accommodation to participants, judges and mentors throughout the event. The event has a total prize money of ₹75,000 for the top three teams to encourage excellence in innovation, implementation and presentation.

The prizes will be awarded as follows:

- **First Prize:** ₹40,000
- **Second Prize:** ₹25,000
- **Third Prize:** ₹10,000

The event takes place on the premises of KJ Somaiya Institute of Management, so there enough facilities with adequate workspace, internet access, constant technical services needed for participants. CodeIcon is a great flagship event for ICON Fest as it aligns with the vision of the institute to promote experiential learning and reinforce its culture of technological excellence.

Hackathon Rules and Guidelines

The following rules will apply to CodeIcon to make sure that everyone is treated fairly, things run smoothly, and everyone has a good time:

1. Team composition:

- There can be 2 to 4 people on each team.
- Before the event, all participants must sign up as a team.
- Once the event starts, no one on the team can change.

2. Problem Statements

- At the beginning of the hackathon, problem statements will be made public.
- Teams can only work on the problem statement they were given or chose.
- If you try to work on problem statements that haven't been approved, you could be disqualified.

3. Rules for Coding and Development

- All development must happen within the 24-hour window.
- Teams can use libraries, frameworks, and APIs that are open to the public.
- You can't use pre-written code, but you can use template setups like boilerplate React, Express, etc. only if the organisers say it's okay.

4. Behaviour and Fair Play

- It is against the rules to copy or plagiarise from other teams.
- You can't get help from outside sources (mentors, friends, or online collaborators) during the event.
- If you break the rules or act inappropriately at the event, you will be disqualified right away.

5. Using Resources

- You can use the internet, but you have to follow the rules of the institute.
- You must use devices responsibly; damaging infrastructure will result in fines.

6. What you need to submit

- Along with a working prototype, teams must also send:
- Code source
- Short documentation
- Demo show
- We will not accept submissions that are late.

7. Managing Your Time

- The hackathon will last for 24 hours straight.
- People who are taking part are expected to take care of their own time, including breaks.

8. How to Use AI Tools

- You can only use AI tools like ChatGPT and Copilot to help you debug or come up with ideas, not to make full solutions.
- You could get in trouble if you rely too much on AI-generated code or use it incorrectly.

9. The Last Presentation

- Every team will give a short demo that lasts 5 to 10 minutes.
- Judges may ask questions to see if the work is original and if the person understands it.

Industry Outreach & Company Collaboration

In advance of planning activities for CodeIcon, a formal outreach process to industry was conducted to secure partners, sponsorship, and invite industry participation for the hackathon. The goal of this outreach was to stimulate substantive industry engagement to improve the quality of the event and academic–industry engagement.

Formal correspondence was initiated with a variety of organisations via email, with sponsorship and collaboration proposals and details such as the purpose of CodeIcon, estimated participant levels, participation levels, and benefits for sponsoring organisations. Additionally, the outreach team sent invitations to specific organisations and industry professionals via LinkedIn to broaden outreach and engage directly with potential sponsors and collaborators.

The goal of this process was simply to invite companies in the field of CodeIcon to engage with the event through sponsorship, guidance of participants, judging panels, resources, or internship possibilities. Outreach is ongoing, and the organising team is continuing to meet with representatives from companies to obtain formal partners in the lead up to the agreed upon dates.

Email template used:

in:sent

Active

SOMAIYA

9 of 1,133

Invitation for Collaboration – “Code Icon” Hackathon (18th–19th December 2025)

**ICON SIMSR** <icon.simsr@somaiya.edu>
to jesvin.varghese

Mon, Nov 17, 5:10 PM (4 days ago)

Dear Ms. Jesvin Varghese,

Greetings!

We are delighted to invite **Fractal Analytics Inc.** to collaborate with us as the **Hackathon Partner** for our upcoming national-level hackathon, “**Code Icon**”, organized by the **MCA Batch 2024–26 of KJ Somaiya Institute of Management, Mumbai**. The event is scheduled for **18–19 December 2025** and aims to promote impactful and real-world technical innovation.

Event Details

Event Name: Code Icon
Format: 24-Hour Offline Hackathon
Dates: 18–19 December 2025
Participants: ~100 Teams (~400 Students)
Prize Pool: ₹80,000
Venue: KJ Somaiya Institute of Management, Mumbai

Collaboration Opportunities

We would be grateful for Fractal Analytics Inc.’s support in any of the following areas:

- Prize sponsorship
- Providing industry-relevant problem statements
- Mentors or judging panel representation
- Certificates and trophies
- Goodies or merchandise for participants

Planning and Preparation

Planning and Preparation was a very important stage that laid the foundation for our initiative in the NEP Hackathon. During this stage, we had a series of meetings with our assigned NEP mentor, where objectives were clarified, the expectations understood, and the general workflow streamlined for the said event. This helped us organize our approach, portion out responsibilities, and ensure each team member was on the same page regarding the project goals.

Developing the appropriate format for a professional email that could be used in outreach was one of the main tasks under this phase. Specifically, the provided template of the email had been developed to help explain the purpose of the hackathon, to enumerate its benefits, and formally request either sponsorship or collaboration. A standardized format helps make communications uniform and professional with potential partner organizations.

In all, it was the planning and preparation phase that provided us with coordination, well-defined roles, and a systematic process to guide us through our work.

Technical Setup & Coordination

In advance of planning activities for CodeIcon, a formal outreach process to industry was conducted to secure partners, sponsorship, and invite industry participation for the hackathon. The goal of this outreach was to stimulate substantive industry engagement to improve the quality of the event and academic–industry engagement.

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Event Execution

Phase 1: Outreach, Registration & Selection (Dec 1 – Dec 14)

Focus: Marketing, Idea Submission, and Quality Control

The primary goal of this phase was to attract quality talent rather than just high numbers. We utilized **Unstop** as our primary platform for registration and submission management.

Date/Timeline	Activity	Execution Details
Dec 01	Registration Live	Registrations opened on Unstop. We simultaneously launched a social media campaign and circulated the event details in WhatsApp groups of various technical colleges across India.
Dec 01 - 12	Idea Submission	Teams were required to submit a PPT detailing their proposed solution based on the released themes. We focused on this "Idea Phase" to ensure teams had a clear execution plan before reaching the venue.
Dec 12	Submission Deadline	The portal was closed for submissions at midnight.
Dec 13 - 14	Screening & Shortlisting	A panel of student organizers and faculty mentors reviewed the proposals. We shortlisted the Best 50 Teams based on the feasibility of their idea and their technical approach, rather than just theoretical concepts.
Dec 14	Finalist Announcement	Acceptance emails were sent to the top 50 teams, confirming their spot in the Grand Finale.

Phase 2: Onboarding & Logistics (Dec 15 – Dec 17)

Focus: Communication, Guidelines & Accommodation Verification

Once the teams were finalized, we shifted our efforts toward operations and logistics, especially for outstation participants.

Communication Channel

- An official **WhatsApp group** was created for all finalist team leads.
- This became the primary channel for quick updates, including:
 1. Rulebook
 2. Technical Guidelines
 3. Code of Conduct

Accommodation Verification

- A dedicated **Accommodation Request Form** was shared with finalists.
- **Policy:** Accommodation was provided *only* to participants residing outside Mumbai.
- **Verification Process:**
 - Students were required to upload valid address proof (Aadhar Card / College ID).
 - Our team manually verified each document to prevent misuse.
 - Only verified outstation candidates were allotted accommodation slots.

Date/Timeline	Activity	Operational Detail

Dec 17 (6PM - 9PM)	Outstation Check-in	Teams traveling from other cities arrived at the campus. A dedicated desk handled their check-in process, verified their original documents, and allocated them to the hostel guest rooms/dormitory setup. This early check-in ensured they were rested before the hackathon began the next morning.
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Phase 3: The Main Event (Dec 18 – Dec 19)

Focus: Implementation, Sustenance, and The 24-Hour Marathon

With outstation students settled and registration complete, the main event kicked off on the morning of the 18th.

Strict Adherence to Qualified Ideas: It was explicitly communicated that teams **must work on the same problem statement and solution** proposed in their qualifying PPT. Pivoting to a completely new idea at the venue was not permitted, ensuring that the 24 hours were dedicated to *execution* and *prototyping* rather than fresh ideation

Time	Activity	Operational Detail
Dec 18, 08:00 AM	Breakfast & Check-in	A wholesome breakfast was served for outstation teams at the hostel mess. Simultaneous check-in began for Mumbai-based teams.
10:00 AM	Inauguration	Welcome address by the Director. The "Code of Conduct" was reiterated, emphasizing that teams must build upon their submitted PPTs.

11:00 AM	HACKING BEGINS	The 24-hour countdown started. Teams immediately began setting up their dev environments to implement their pre-approved ideas.
01:00 PM	Lunch Break	Buffet lunch served in slots (Group A & Group B) to prevent overcrowding at the food counters while ensuring work continuity.
03:00 PM	Mentoring Round 1	Implementation Check: Mentors visited tables to verify that the code being written aligned with the proposal submitted during the selection phase.
05:30 PM	High Tea & Snacks	Evening tea with heavy snacks (samosas/sandwiches) was distributed directly to the tables to minimize distraction.
08:30 PM	Dinner	A nutritious dinner was served. Post-dinner, teams moved into the "Deep Work" phase for backend integration.
11:00 PM	Mentoring Round 2	Progress Check: Review of the MVP (Minimum Viable Product). Mentors helped troubleshoot critical bugs.
Dec 19, 01:00 AM	Midnight Energizers	To combat the "graveyard shift" fatigue, we distributed energy drinks and coffee.
03:00 AM	Rest & Rotations	Outstation students utilized the dormitory facility in shifts for power naps, while one member per team stayed active to monitor builds.

08:00 AM	Breakfast	Hot breakfast (Idli/Poha) and fresh coffee were served to wake up the tired participants for the final sprint.
10:00 AM	Mentoring Round 3	Final Polish: Mentors helped teams refine their pitch decks and UI/UX before the code freeze.
11:00 AM	CODE FREEZE	Stop Hacking. All GitHub repositories were locked.
11:30 AM	Judging (Round 1)	Elimination round at the tables. Judges shortlisted the Top 10 teams.
01:00 PM	Lunch	A quick lunch was served for all participants and judges before the final ceremony.
02:00 PM	Grand Finale	Top 10 teams pitched their live prototypes to the main jury on stage.
04:00 PM	Valedictory	Prize distribution and Vote of Thanks.
05:00 PM	Event Conclusion	Local teams departed.
06:00 PM	Extended Stay (Outstation)	Safety Protocol: Recognizing the fatigue of the participants, the institute provided an extra night's stay (Night of Dec 19) for outstation teams to ensure they

		could rest properly before travelling back home the next morning.
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NEP Student Individual Hackathon Participation

This is the **NEP student individual hackathon participation** by Mohammed Varaliya and Vraj Shah:

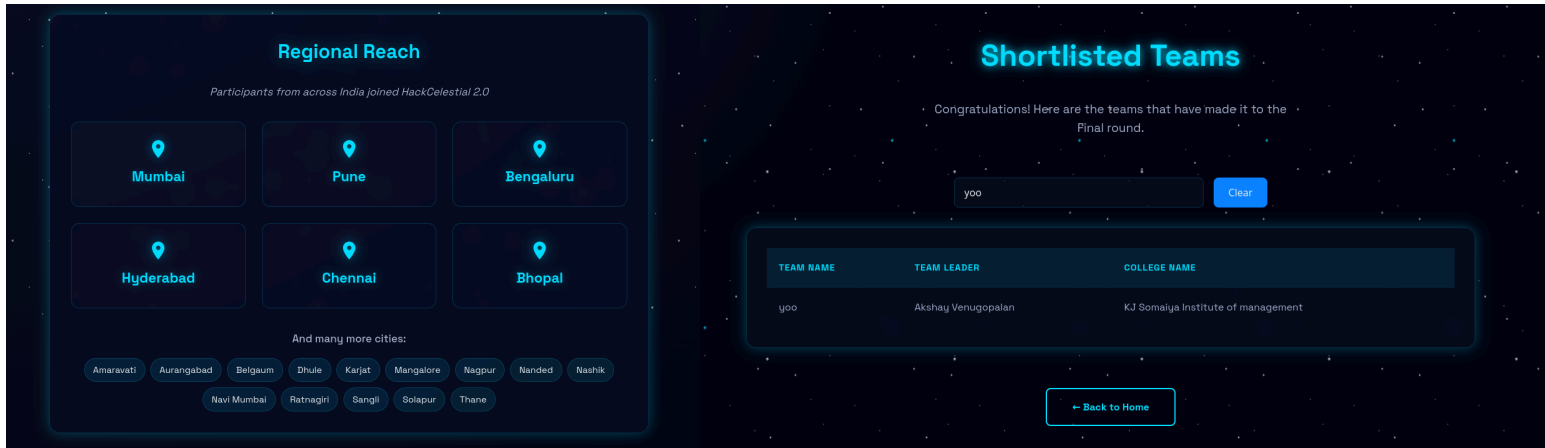
We took part in *The Multilingual Hackathon* (13 Nov – 16 Nov), which was all about “Build anything. Translate everything,” powered by Lingo. [WeMakeDevs+1](#)

During the hackathon, we built a project that’s really close to our hearts, our **Multi-Lingual Prescription Reader**, and here’s the GitHub link: [Multi-Lingual Prescription App](#). [GitHub](#)

What we did & why it mattered:

- We built a web application that lets you upload a prescription image (yes, even handwritten ones), and then it uses AI (Google Gemini Vision) to read and extract the medical information from it.
- Once the prescription is parsed into a structured JSON format, users can review or correct the extracted details, we didn’t want “black-box AI”: we added an editable interface so that real-world mistakes can be fixed.
- After that, the app generates a patient-friendly medical summary, like what each medicine is for, side effects, how to take it, and *then* translates that summary into multiple languages using Lingo.dev’s SDK.
- On top of that, there’s audio playback, so you can **listen** to the summary in your language.
- All this is built with a clean tech stack: React + TypeScript + Vite + Tailwind CSS on the front end, and Node.js + Express for the backend.

This is the **NEP student individual hackathon participation** by Benjamin Sebastian, Akshay Venugopalan, Shaun Rodrigues, Soham Tare:



We took part in *HackCelestial 2.0* (19 Sept – 20 Sept), in which out of 600 teams registered our team qualified one among 72 selected teams.

The problem statement given to us was to build a “one nation one athlete platform” for linking all athletes to connect and interact with coaches, governmental organizations and get connected about match information.

During the hackathon, we as a team in a time span of 36 hours we went through a rollercoaster of changes in our project .

Github: [link](#)





Our Experience:

- We started our project by deciding two of our teammates will handle the frontend and the other two will be handling the backend.
- We worked together seamlessly and throughout the 36 hours we created the entire project from scratch.
- We were told to bring our prototype but we were confident that we will do it all the 36 hours
- We came early 1 day prior for doing the system design and environment setup
- In the 36 hours we used our resources to create a full fledged working application.
- the last 12 hours were especially tough since we didn't sleep for 2 nights in a row



This is the **NEP student individual hackathon participation** by Arunima Pandey.

I took part in Innovesite Hackathon (11 Oct– 12 Oct), which was all about “4E Framework - Education, Experience, Engagement and Exposure” powered by BNP Paribas.

During the hackathon, I built a project that's related to FinTech.

Problem Statement : AML systems protect the finance industry by flagging suspicious behaviors such as unusual transfers, structuring, or attempting transactions with high-risk individuals or entities. Traditional rule-based systems assess a transaction independently of others, allowing a criminal to conceal illicit activity across multiple accounts or multiple layers of transactions. This may result in missed detections, if even one transaction does not trigger the detection, and excessive alerts.

By creating networks out of the transaction data to highlight accounts and relationships, a network analysis can uncover obscure relationships, anomalous clusters, and complex patterns of money laundering that are harder to uncover with traditional systems.

What I did & why it mattered:

- Developed a web-based AML system that represents raw transactions into a network of accounts and relationships.
- Modeled accounts as nodes with both transaction edges and intrinsic connections such as shared IDs or ownership.
- Built an interactive graph visualization of the network to allow users to explore flows, filter data, switch layouts, and view attributes.
- Incorporated an anomaly detection system that uses graph features and embeddings to automatically flag accounts as suspicious.
- Created an Alerts dashboard so investigators can click an alert and instantly see the surrounding network.
- Assisted in identifying complex laundering patterns that traditional rule-based systems typically miss.

Challenges Faced & Solutions

No event of this scale is without hiccups. We faced several challenges throughout the process but managed to resolve them through quick thinking and adaptability.

1. Accommodation for Outstation Teams

- **The Challenge:** We had a larger-than-expected turnout of participants from outside Mumbai, which put a significant strain on our initial housing plans.
- **Our Solution:** The Logistics Team acted quickly, coordinating with the hostel warden to utilize vacant guest rooms. For the overflow, we converted the institute's common activity hall into a temporary "sleeping zone" equipped with mattresses and fresh linens, creating a safe and budget-friendly on-campus stay.

2. Securing Sponsorships

- **The Challenge:** One of the toughest hurdles was the initial lack of response from corporate sponsors. Cold outreach often hit dead ends, and many companies had already closed their budgets for the year.
- **Our Solution:** We pivoted our strategy by leveraging our alumni network for warmer introductions. We also broadened our scope to accept "in-kind" sponsorships (such as server credits, energy drinks, and merchandise) rather than just demanding cash. This secured necessary resources without straining our internal budget.

3. Wi-Fi Bandwidth Issues

- **The Challenge:** About six hours into the coding phase, the network slowed down significantly due to heavy traffic.
- **Our Solution:** The technical team immediately set up a secondary backup line and restricted non-essential streaming on the main network to prioritize GitHub pushes and deployments.

4. Power Load Management

- **The Challenge:** With hundreds of laptops charging simultaneously, we faced a risk of tripping the circuit breakers.
- **Our Solution:** We distributed the power load across different phases of the building's electrical grid and monitored usage strictly to prevent outages.

5. Judging Delays

- **The Challenge:** The evaluation process took longer than expected, threatening to push the closing ceremony late into the evening.
- **Our Solution:** We split the judging panel into parallel streams, allowing two teams to be evaluated simultaneously. This doubled our speed and brought us back on schedule.

6. Participant Fatigue

- **The Challenge:** By 3:00 AM, energy levels among participants dropped significantly.
- **Our Solution:** The organizing team initiated "energizer" gaming breaks and distributed fresh coffee and snacks to keep morale high.

Financial Overview

Percentage may be vary (rough estimation)

Category	Description	Approx. Allocation (%)
Prize Pool	Cash awards for Top 3 teams	40%
Logistics	Venue setup, sound, internet, stationery	20%
Food & Beverage	4 meals + snacks for 200+ people	25%
Marketing/Swag	Banners, T-shirts, ID cards, certificates	10%
Miscellaneous	Contingency fund for emergencies	5%

Conclusion

The NEP Hackathon (CodeIcon) is more than just a competition; it is a comprehensive exercise in management, technology, and collaboration. For the participants, it was a test of technical grit. For us, the NEP student organizers, it is a real-world lesson in how to turn a vision into a reality.

We successfully navigated the complexities of industry outreach, logistical planning, and crisis management. We hope that the event aligns perfectly with the National Education Policy's vision by fostering innovation and providing a platform for experiential learning. As we move forward, the lessons that we learn here specifically regarding teamwork and technical adaptability will serve as a strong foundation for our future academic and professional endeavors.